

Computational & Data Sciences

Indian Institute of Science, Bangalore

AUGUST–DECEMBER SEMESTER, 2025

DS290 MODELING AND SIMULATION

Final Examination

Time allowed: 180 minutes

NOTE: Release: 24 Nov 2025, 1400 hours. Due by: 24 Nov 2025, 1700 hours.

Points are uniformly distributed over the questions. There are 8 questions, each worth $\frac{25}{4}$ points. **Full points: 50.**

IMPORTANT: Please write your Name-

SR No.-

the approximate number of lectures (out of 24) you estimate you attended.

1. Minimize the number of noise channels (components of the Wiener process) in the n -dimensional Itô stochastic differential equation

$$dX_t = a dt + (I_{n \times n} - UU^\top) dW_t$$

where W_t is an n -dimensional Wiener process, $U \in \mathbb{R}^{n \times k}$, $k < n$, with $U^\top U = I_{k \times k}$, and $a \in \mathbb{R}^n$ is constant. Find the reduced-noise SDE with the same diffusion properties.

2. Let $\{W_t^{(i)}\}_{t \geq 0}$ for $i = 1, \dots, m$ be m independent n -dimensional Brownian motions, each with independent components. Let $U_i \in \mathbb{R}^{n \times n}$ be m constant unitary matrices. Find the constant a such that

$$\frac{1}{a} \sum_{i=1}^m U_i W_t^{(i)}$$

is also a Brownian motion. Use matrix notation only.

3. Consider the numerical method:

$$X_{n+1} = X_n + X_n \left(1 + \frac{1}{2} \Delta W_n\right) \Delta W_n$$

for approximating the SDE

$$dX_t = X_t dW_t, \quad t \in [0, T],$$

where ΔW_n is the Wiener increment with step size h .

- (a) What multi-indices from $\{0, 1\}$ are used? Identify the hierarchical and remainder sets.
(b) What is the expected drift term from the method?
(c) Does the method converge (strongly/weakly)? If no, identify which term causes non-convergence.
4. Consider the SDE system:

$$dX_t^{(1)} = dW_t^{(1)}, \quad dX_t^{(2)} = X_t^{(1)} dW_t^{(2)},$$

and the numerical scheme:

$$X_{n+1}^{(1)} = X_n^{(1)} + \Delta W_n^{(1)},$$
$$X_{n+1}^{(2)} = X_n^{(2)} + X_n^{(1)} \Delta W_n^{(2)} + \frac{\Delta W_n^{(1)} \Delta W_n^{(2)}}{2} - \frac{V_n}{2},$$

where $P(V_n = \pm \Delta_n) = \frac{1}{2}$.

Estimate the strong and weak order of convergence for $X^{(2)}$.

5. Applying the Milstein method to the SDE

$$dX_t = \sqrt{|X_t|} dW_t,$$

find the **largest time step** h such that, with probability 1,

$$\mathbb{E}[(\delta X_1)^2] \leq \mathbb{E}[(\delta X_0)^2],$$

where δX_1 is the perturbation after one Milstein step.

6. For the SDE

$$dY_t = \begin{bmatrix} 0 \\ 1 \end{bmatrix} dt + \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} dW_t^{(1)} \\ dW_t^{(2)} \end{bmatrix},$$

find the Girsanov transform $\widetilde{W}_t$ and the probability measure Q such that under Q ,

$$d\widetilde{Y}_t = \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} d\widetilde{W}_t^{(1)} \\ d\widetilde{W}_t^{(2)} \end{bmatrix}.$$

7. (a) Find the expected Lyapunov exponent (using $|X_t|$) for the SDE

$$dX_t = -X_t(2 \ln X_t + 1) dt - 2X_t \sqrt{-\ln X_t} dW_t,$$

with $X_t \in (0, 1]$, $X_0 = e^{-1}$. Is the zero solution asymptotically stable in the mean?
(Hint: let $u_t = \sqrt{-\ln X_t}$.)

(b) For the SDE

$$dX_t = f(X_t) dt + \sum_{j=1}^m b_j(X_t) dW_t^{(j)},$$

find the Itô form of the Stratonovich integral

$$\sum_{j=1}^m \int_0^t g_j(X_s) \circ dW_s^{(j)}.$$

8. Let $p_b(X_t = x \mid X_T = a)$ be the backward transition density for the SDE

$$dX_t = f(X_t) dt + \sum_{j=1}^m b_j(X_t) dW_t^{(j)}.$$

Suppose the observation process is

$$dY_t = h(X_t, t) dt + dW_t,$$

with $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Find an $h(X_t, t)$ such that the *observed* backward transition density

$$\hat{p}_b(X_t = x \mid X_T = a)$$

is constant. State all assumptions clearly.